Hamza HOUIR ALAMI

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Portfolio | LinkedIn

Profile & Objectives

Industrial and Mechanical Engineering student at Arts et Métiers, with interests in operations research, optimization, manufacturing systems, and artificial intelligence.

Education

Combined undergraduate GPA: 3.89/4.0

Arts et Métiers Institute of Technology — Lille / Paris, France
Combined B.S. and M.S. in Industrial and Mechanical Engineering — *Sept 2025 – Expected Jun 2028*
Arts et Métiers is part of ParisTech, a consortium of prestigious French higher education institutions known for its high standard of education and the excellence of its teaching staff and laboratories.

- Top 10 French Graduate Engineering School.
- Relevant courses : mathematics, computer science, machine learning, optimization, production planning, and manufacturing systems.

Lycée Faidherbe — Lille, France
Intensive Preparatory Program for Competitive Admission to Elite Engineering Schools — *Sept 2023 – Jul 2025*

- Top 10 selective two-year program in France.
- Relevant courses : Linear Algebra, Calculus, Probability, Differential Equations, Numerical Methods, Mechanics, Thermodynamics, and Electromagnetism.

Work Experience

Mohammed VI Polytechnic University – IT Department — Rabat, Morocco
AI & Operations Research Intern — *Jul – Aug 2026*

- Developed a Python/FastAPI decision-support tool to allocate IT equipment across teams and labs, adopted by the IT Department.
- Benchmarked 9 demand-forecasting models on 60 series with rolling-origin validation to drive replenishment decisions.
- Designed inventory and ordering policies that cut simulated cost by 21% and emergency purchases by 92% at a 94% service level.

Safran Nacelles — Casablanca, Morocco
Operations & Manufacturing Engineering Intern — *Jun – Jul 2026*

- Optimized routing time on a nacelle assembly line with quadratic programming and Monte Carlo simulation, meeting the reduction target.
- Conducted an R&D feasibility study on radiant heating to accelerate epoxy adhesive curing, modeling thermal gradients and cure kinetics.
- Redesigned assembly tooling in CAD to fit flow and access constraints.

AMJE Lille – Arts et Métiers Junior-Entreprise — Lille, France
Head of Business Development & Partnerships — *Dec 2025 – Present*

- Leading 15 business managers; coordinating client qualification, proposals, commercial follow-up, staffing needs, and partnership development for a student-run consulting firm.
- Grew signed revenue by 85% since May 2026 (EUR 27K → 50K, 10 contracts), supported by expanding the prospecting database from approximately 600 to 1,600 contacts.

Decathlon — Remote, Part-time
Consultant via AMJE Lille — *Jul 2026 – Present*

- Delivered a client engagement focused on product data structuring; project details confidential.

Academic and Research Projects

Compositional Spoken-Number Recognition from Synthetic Speech (Preprint, Nov. 2026) — 2026
Collaborative course project on English digits; independent extension to French, English and Arabic numbers

- Built a log-Mel CNN for English digit recognition; leave-one-speaker-out evaluation showed **65%** mean accuracy on unseen speakers vs. 95% on known speakers.
- Recognized French numbers **0–99** without any real speech or compound number in training: a CTC head on frozen Whisper features learns from **23 synthetic (TTS) base words** with filtered cross-fade augmentation, and grammar-constrained decoding composes the numbers.
- Achieved **86.7%** accuracy on 928 real recordings from 45 speakers, including **77.3%** on compound numbers vs. 31.3% for an isolated-word baseline; quantified uncertainty with speaker-level bootstrap intervals and evaluated noise robustness.

Flywheel Rotor Optimization 2024 – 2025

Supervised research project

- Proved a perimeter bound with Wirtinger's inequality and separated inertia from regenerative-braking models.
- Optimized rotating-disk profiles for energy per unit mass; slope-limited designs reach 2.2× a uniform disk in 2D analysis.
- Pre-registered hash-frozen predictions and an uncertainty-budgeted test protocol for 3D-printed rotors.

Shoulder Prosthesis Design and Optimization Sept 2026 – Present

Five-member research project in collaboration with the Georges Charpak Institute of Human Biomechanics (Arts et Métiers)

- Investigating glenoid implant loosening through bone–implant contact modeling in Abaqus and 3DEXPERIENCE.
- Optimizing implant geometry, surface patterns and materials to improve fixation strength and durability.

Glider Wing – Aerodynamic and Structural Study Fall 2025

Supervised two-person engineering project with LISPEN Research Laboratory (Arts et Métiers Lille)

- Benchmarked airfoils and ran 2D CFD in STAR-CCM+ (inviscid, laminar, k-ϵ turbulent) with mesh convergence.
- Sized the spar via a Python material sweep and Abaqus FEA, selecting Al 7075-T6 for minimum mass.

DeskBuddy – Autonomous Desktop Cleaning Robot Oct 2026 – Present

Independent Mechatronics Design Project : First prototype in preparation.

- Designing a desk-cleaning robot combining compact suction and edge detection with a conversational AI head (voice, animated eyes).
- Defined the robot architecture for 20–30 min runtime; modeled 26 3D-printed parts in CAD.

Formula Student – FSAM, Arts et Métiers Lille 2025 – 2026

Bill of Materials and Inventory Coordination

- Structured a vehicle bill of materials across **18+ assemblies** in Excel and 3DEXPERIENCE; reconciled required parts with workshop inventory, suppliers, and technical-subteam needs.

Leadership & Community Engagement

Student Association of Arts et Métiers Lille Lille, France

Business Relations Officer *Apr 2026 – Present*

- Organizing employer workshops, networking events, and industrial-site visits.
- Maintaining relationships with companies including Hermès, Safran, EDF, and L'Oréal.

Science Outreach Program Paris, France

Team Member *2026 – Present*

- Collaborating in an eight-member team to design hands-on physics experiments and film demonstrations for children.

Mathematics and Physics Tutor 2024 – Present

Have provided individualized tutoring to **10 students**, addressing learning difficulties and preparing them for examinations.

Mentor for Moroccan Students 2026 – Present

Guiding **5 students** through French engineering education pathways and the transition to studying in France.

Competitive Football 13 years

Served as team captain and competed in national youth tournaments and the 2018 IberCup Cascais in Portugal.

Awards & Achievements

Genin Scholarship (2025)

Scholarship supporting academic excellence and integration of international students in France.

FC Barcelona Escola Diploma

Completed the academy's youth football training program.

Football Titles

Won a Casablanca regional tournament and the Lycée Faidherbe interclass tournament.

Technical Skills & Languages

Programming and data: Python, SQL, pandas, NumPy, SciPy, scikit-learn, PostgreSQL, Excel.

Engineering tools: CATIA/3DEXPERIENCE, Abaqus, STAR-CCM+; CAD, finite-element analysis, and CFD.

Languages: Arabic (native), French (native), English (advanced).

Interests: Formula 1, Football and geopolitics.